Contact-Aligned Local Graphs with Tricubic Signed-Distance Refinement for Smooth Rigid Frictional Contact

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Abstract

Rigid frictional contact requires a signed gap, contact normal, and tangential frame that vary smoothly as bodies slide or roll. When these quantities are obtained from faceted collision proxies or fixed surface tessellations, contact-point motion across cells can produce nonphysical friction-force spikes and stick–slip chatter. We propose Contact-Aligned Local Graphs with configuration-space signed-distance refinement (CALG-SDF), a contact-geometry backend for smooth rigid frictional contact. CALG constructs a local contact frame for each broad-phase candidate and, when a normal-cone certificate is satisfied, represents the two patches as height graphs over a shared contact plane. The resulting signed gap samples a local configuration-space signed-distance field queried by tensor-product cubic interpolation; the rigid response uses the interpolated gap and gradient normal. Analytic inclined-plane controls first verify the implemented static and kinetic Coulomb response. Four same-scene dynamic benchmarks then compare CALG-SDF with two triangle-based baselines, Linear-triangle BVH and PN-triangle BVH, under identical rigid-body states and friction parameters. Linear-triangle BVH produces 71.9–2.20 $\times$ 10^3 times larger contact-normal angle jumps and 37.7–108 times larger friction-force jumps than CALG; PN-triangle BVH produces 40.2–826 times larger normal-angle jumps and 35.8–62.6 times larger friction-force jumps. Comparisons with the independent multibody solvers Chrono and Recurdyn show trajectory-level agreement in displacement and velocity trends in three selected engineering scenes, and solver-loop timings are reported for the implemented validation benchmark.

Keywords: Frictional contact, rigid-body dynamics, local graph representation, signed distance fields, tensor-product cubic interpolation, contact detection, computational contact mechanics

1. Introduction

Rigid frictional contact requires a contact solver to repeatedly provide a signed gap, a contact normal, and a tangential frame as bodies slide, roll, and impact. These quantities are not auxiliary outputs: normal response is activated from the signed gap and normal direction in classical contact mechanics [1, 3], while Coulomb friction projects relative motion onto the tangent plane and evolves a discrete stick–slip state [2, 4]. In rolling bearings, cam–roller followers, guide slots, and ball-and-socket joints, small discontinuities in these geometric inputs can therefore appear as friction-force spikes, tangential-speed jitter, stick–slip chatter, and acceleration-level artifacts. This makes the contact-geometry backend part of the friction model rather than a passive collision-detection utility.

Most practical simulations decouple the mechanics update from the contact geometry through collision proxies or fixed surface tessellations. A Linear-triangle BVH is efficient and robust for broad use, but its face-based gap and normal are only piecewise smooth. PN-triangle BVH and fixed tessellations of higher-order surfaces provide smoother local geometry, yet the returned contact frame can still depend on the chosen tessellation and on how the contact point crosses cells. Smooth surface representations reduce many artifacts of piecewise C^0 contact geometry [8], and high-order contact formulations improve geometric accuracy in elastodynamic contact [9]; however, dense linear proxies are still commonly used for collision and force transfer. Global refinement, global signed-distance fields, or global surface parameterizations can reduce some discretization effects, but they increase broad-phase work, memory, or topology management and do not by themselves make the friction frame a controlled local object. The unresolved issue is therefore how to obtain smooth contact-frame quantities without relying on global refinement or a global parameterization.

This paper proposes Contact-Aligned Local Graphs with configuration-space signed-distance refinement (CALG–SDF) for this setting. The key observation is that contact itself supplies the missing local coordinates. For each broad-phase candidate, CALG builds a contact frame from preliminary closest-point and normal information; when a normal-cone graphability certificate is satisfied, the two local patches are represented as height graphs over a shared contact plane. The signed gap, normal, and tangential basis are then evaluated in this contact-aligned frame, and non-graphable configurations are routed to a generic curved patch-patch evaluation or a conservative fallback. This converts the near-phase query from one governed by fixed tessellation topology into a certified local graph query with an explicit validity boundary.

The second layer turns these local samples into a response object suitable for repeated rigid-body friction evaluations. In the implemented backend, CALG contact samples define a configuration-space signed-distance response field for the rigid contact element, and tensor-product cubic interpolation provides the signed gap and gradient normal during the solver loop. This SDF layer does not replace the geometric construction; it makes the certified local contact samples reusable at solver-loop cost. The resulting backend treats signed gap, normal, and tangent frame as coupled geometric inputs to friction, reducing the geometric source of force spikes and acceleration artifacts while retaining explicit fallback control.

We evaluate this mechanism through a sequence of tests that separates friction correctness, geometry-backend behavior, external-solver agreement, and solver-loop timing. Analytic inclined-plane controls verify the implemented static and kinetic Coulomb response. Four same-scene dynamic friction benchmarks then compare CALG–SDF with Linear-triangle BVH and PN-triangle BVH under identical rigid-body states and friction parameters. Three selected engineering scenes are also compared with Chrono and Recurdyn to assess trajectory-level agreement in displacement and velocity trends, and solver-loop timing reports the implemented backend cost. These results support a bounded claim: CALG–SDF is a contact-geometry backend for rigid frictional contact that improves contact-frame continuity and practical solver-loop efficiency in the tested scenes.

The proposed CALG–SDF formulation makes four contributions.

(i) *Contact-aligned graph representation of local contact geometry.*

CALG constructs a contact-defined local frame in which graphable contact patches are represented as height graphs over a shared contact plane. This representation provides contact-frame signed gaps, normals, and tangent bases that are governed by local smooth geometry rather than by fixed tessellation topology.

(ii) *Configuration-space response reconstruction from local graph samples.*

Local CALG contact samples are converted into a configuration-space signed-distance response field for rigid contact elements, which is queried by tensor-product cubic interpolation during repeated frictional response evaluations. This converts repeated contact-response queries into smooth local field evaluations of signed gap and gradient normal.

(iii) *Friction-oriented continuity of contact-frame quantities.*

The formulation treats signed gap, contact normal, and tangential frame as coupled geometric inputs to Coulomb friction. This makes contact-frame continuity a primary design objective and reduces the geometric source of friction-force spikes, tangential-speed jitter, stick–slip chatter, and acceleration-level artifacts.

(iv) *Graphability-certified reduction with explicit validity control.*

A normal-cone graphability certificate determines when the local height-graph reduction is valid, and non-graphable cases switch to generic curved patch-patch evaluation or conservative fallback. This gives the backend an explicit validity boundary instead of applying a local graph approximation outside its geometric regime.

The remainder of the paper follows this argument. Section 2 reviews rigid-body frictional contact, faceted triangle-based contact proxies, curved and smooth contact geometry, signed-distance contact fields, and local contact-chart alternatives, with emphasis on where the contact-geometry backend enters frictional response. Section 3 presents the CALG detector, graphability certificate, response-field construction, configuration-space SDF refinement, error estimates, and fallback hierarchy. Section 4 first verifies the friction law with analytic inclined-plane controls, then isolates the contact-geometry module through same-scene CALG–SDF, PN-triangle BVH, and Linear-triangle BVH dynamic comparisons. The same section then reports Chrono–Recurdyn–CALG external-validation scenes and solver-loop timing evidence before summarizing the scope and limitations of the current rigid-contact backend.

2. Related work

This work concerns the geometric backend of rigid-body frictional contact. The relevant literature is therefore organized around how contact solvers obtain gap values, contact normals, tangent frames, and smooth signed responses, rather than around finite element contact enforcement alone. Classical contact mechanics established variational contact and frictional constraints [1, 2]. Computational treatments then systematized penalty, multiplier, and augmented Lagrangian contact enforcement [3, 4]. These developments provide the mechanical background, but the present paper focuses on the geometry module that supplies contact data to a rigid-body frictional solver.

2.1. Rigid-body frictional contact and multibody dynamics

Rigid-body contact with Coulomb friction is commonly advanced through time-stepping complementarity, nonsmooth dynamics, or compliant regularization. Implicit time-stepping and complementarity formulations provide stable updates for inelastic impacts and contact among multiple rigid bodies [13, 14]. Nonsmooth contact dynamics and its numerical analysis give another route for unilateral constraints with friction [15, 16]. Semi-implicit and geometrically implicit schemes target convergence and intermittent contact [17, 20]. Cone-complementarity and matrix-free formulations address large rigid-body systems [18, 19].

Compliant contact models are widely used in multibody solvers because they regularize impact and friction into force laws that are practical for engineering dynamics. Machado et al. review compliant force models derived from Hertz-type contact theory [21], and recent reviews summarize contact phenomena in dynamical and multibody systems [22, 23]. Project Chrono is used here as an independent multibody reference because it provides a documented open-source implementation of rigid and multiphysics contact simulation [24]. These works primarily address time integration, complementarity, regularization, or solver implementation. They still require a geometric backend that returns reliable gaps, normals, and tangent frames; errors or discontinuities in these quantities enter the friction law directly.

2.2. Faceted collision proxies and triangle-based BVH contact

Most practical contact pipelines separate broad-phase candidate generation from narrow-phase distance or intersection queries. GJK distance queries [25], OBB trees [26], and V-Clip for polyhedral proximity [27] are representative building blocks for rigid collision detection. Deformable and animation-oriented collision pipelines further combine spatial hierarchies, continuous collision detection, and robust contact handling [28, 29]. Geometrically exact CCD methods reduce missed collisions for moving primitives [30], but the contact response is still often evaluated on a faceted or tessellated proxy.

The Linear-triangle BVH baseline studied in this paper belongs to this faceted-proxy family. Its attraction is clear: triangle BVHs are simple, robust, and fast to traverse. Its weakness is also geometric: the contact normal and tangent frame are piecewise constant or piecewise reconstructed from adjacent triangles, so a sliding contact point can experience abrupt frame changes when it crosses element boundaries. PN-triangle BVH is a smoother direct baseline because it reconstructs a curved local proxy from triangle data [42], but its contact frame still depends on local cell reconstruction and tessellation. In frictional contact this is not a cosmetic error. The normal defines the tangential projection, the tangent frame defines the friction direction, and both affect force spikes, tangential-speed jitter, and stick–slip chatter.

2.3. Curved and smooth contact geometry

Curved and smooth contact representations address the geometric discontinuity of faceted proxies. PN triangles provide a compact curved surface reconstruction from vertex positions and normals [42]. Smooth surface representations have recently been shown to reduce tangential force artifacts in deformable contact [8], while high-order IPC improves contact accuracy on curved meshes [9]. Isogeometric contact analysis provides another route to smooth geometry by using NURBS-based surfaces for contact treatment [31, 32]. Large-deformation frictional isogeometric contact and review work further clarify how exact or high-order geometry can improve contact discretization [33, 34]. These methods demonstrate the value of smooth geometry, but most are designed for deformable or high-order discretizations; the present work uses this geometric motivation in a rigid frictional contact backend.

Variational and potential-based contact methods also motivate a smoother geometric interface. IPC embeds non-penetration in a barrier potential for implicitly time-stepped dynamics [6]. Surface-potential contact integrates interactions over smooth surfaces rather than relying only on isolated point constraints [5], and Geometric Contact Potential formulates contact potentials from continuum smooth or piecewise smooth surfaces [7]. Continuous collision detection between parametric surfaces shows a related need for certified queries on curved geometry [10]. These methods show that contact response benefits when the geometry is smoother than a raw triangle proxy. The distinction of CALG is that it targets the local geometry backend itself: broad-phase candidates remain close to the original surface representation, while the narrow phase builds a contact-aligned gap field and frame for frictional rigid dynamics.

2.4. Signed-distance and offset-based contact fields

Signed-distance fields and implicit surfaces provide continuous spatial queries for collision and contact. Level-set methods established signed implicit fronts for geometric evolution [35], and volumetric reconstruction work popularized distance-like fields for complex scanned geometry [36]. Surveys of three-dimensional distance fields describe their use in geometry processing, visualization, and collision applications [37]. More recent contact work has used volumetric offsets and SDFs to improve contact robustness; Offset Geometric Contact constructs offset shapes with orthogonal contact forces [11], and Triangle-SDF CCD casts time of impact as a local spatio-temporal optimization [12].

The SDF component in this paper has a narrower role than a full volumetric collision engine. The primary search is still surface based: CALG supplies candidate pairs, local graph certificates, fallback decisions, and contact samples. For the rigid validation backend, these outputs are coupled to a local configuration-space signed-distance refinement that updates the response gap and normal. This separation matters for the method claim. CALG determines where the local contact geometry is valid, whereas the signed field provides smooth force-activation and friction-response quantities once a candidate has been identified.

2.5. Local contact charts and contact-aligned geometric backends

Surface parameterization and local charts are central tools for representing surface quantities in geometry processing. Least-squares conformal maps introduced a practical conformal atlas construction for mesh parameterization [38], and surveys summarize parameterization methods and their applications [39]. Mean value coordinates provide smooth interpolation over planar polygons [40]. Boundary First Flattening offers boundary-controlled conformal flattening for disk-topology meshes [41]. These methods are useful for global or persistent surface coordinates, but global parameterizations require cuts, topology management, chart overlap handling, and seam treatment. They solve surface representation, whereas frictional contact requires a transient chart tied to the active contact frame and response law.

CALG instead constructs local contact-aligned coordinates on demand for each candidate pair. The local graph gap, normal-cone graphability certificate, and fallback path are designed so that regular smooth contact can be handled by a low-dimensional gap representation, while difficult cases can fall back to conservative curved-patch queries. In this sense, CALG is neither a global parameterization method nor a standalone dynamics solver. It is a contact-geometry backend that supplies smooth signed gaps, normals, tangent frames, and fallback decisions to a rigid-body frictional response model.

Taken together, the prior work shows that rigid frictional dynamics depends strongly on the geometric quantities supplied by collision detection, that faceted proxies can introduce frame discontinuities, and that smooth surfaces or signed-distance fields can reduce such artifacts when they are integrated into the contact pipeline. The remaining gap is a practical backend that keeps the efficiency and candidate-generation simplicity of triangle-based detection while producing smooth contact-frame data suitable for frictional rigid-body dynamics. CALG is designed for this role: it couples local curved-graph contact detection, graphability-aware fallback, and configuration-space signed-gap refinement into a unified geometry backend evaluated in the rigid friction scenarios below.

3. Method

3.1. Problem setup and backend overview

The proposed method is a contact-geometry backend for rigid frictional response. Given two boundary surfaces $S_A, S_B \subset \mathbb{R}^3$ and a rigid configuration of body B, the backend returns the geometric quantities that enter a Coulomb-type contact step: signed gap, contact point or reduced response coordinate, contact normal, tangential frame, and a certificate describing how the query was resolved. The input surface may be analytic, CAD-like, or triangulated with normals and local smoothness information.

The construction follows a standard logic from computational contact mechanics: a mechanical contact law first needs a signed separation and a contact frame, and any discontinuity in these geometric quantities is transferred directly to normal and frictional response [3, 4]. The geometric part is therefore separated into two layers. The first layer, CALG, is a local detector based on differential surface geometry. It builds a contact-aligned frame for each candidate pair, proves when the two patches can be written as height graphs over the same contact plane, evaluates a signed graph gap in that chart, and falls back to a generic curved patch-patch solve when the graph condition is not valid. The second layer is a rigid response field. It tabulates the signed gap generated by the detector in a local configuration-space chart and evaluates the repeated rigid response by tensor-product cubic interpolation. The theoretical sources are consequently explicit: local graph reduction follows from the inverse function theorem for regular surfaces, conservative acceptance and rejection follow from normal-cone and interval-style geometric bounds, and the response field follows the signed-distance and level-set view of contact geometry [35, 37].

For a rigid configuration $q = (x, R) \in SE(3)$, the configuration-space signed gap is written as

$$\phi(q) = g_{\mathrm{CALG}}(S_A,\ RS_B + x), \tag{1}$$

where g_{CALG} is the signed gap returned by the certified detector path, S_A is the fixed or reference contact surface, and $RS_B + x$ is the moved rigid contact element. A general contact element may be orientation dependent; the response field is therefore represented in a local chart of $SE(3)$ and queried through the same full-pose contract for all rigid contact elements.

Table 1: Method modules and their theoretical role. The table is used to separate the detector, the response field, and the fallback hierarchy so that each approximation has a defined validity boundary.

Module	Input and output	Role in the derivation
Contact element	Analytic patch, CAD patch, curved mesh patch, or rigid mesh element; outputs local curved jets and optional surface velocity	Provides a common representation for arbitrary rigid contact features without putting scene-specific cases in the solver loop
Conservative broad phase	Inflated primitive boxes and rigid motion bounds; outputs candidate primitive pairs	Preserves the efficiency of three-dimensional candidate generation while keeping geometric error and motion bounds explicit
Contact-aligned graph detector	Candidate curved primitives; outputs signed samples, normals, tangent frames, and certificates	Converts regular smooth contact into a local height-graph gap when the normal-cone certificate proves that the projection is valid
Generic fallback	Non-graphable or uncertain candidates; outputs a certified closest-patch or interval/subdivision result	Covers open surfaces, boundaries, sharp features, and ill-conditioned local charts without accepting an invalid graph approximation
Configuration-space response field	Certified detector samples in a local $SE(3)$ band; outputs interpolated signed gap and gradient normal	Converts repeated rigid response queries into smooth tensor-product cubic field evaluations while retaining detector fallback outside the valid band

The pipeline is designed around five requirements. First, it is local: neither layer requires a global surface chart, user-specified cut, or seam handling. Second, it is certificate driven: smooth curved geometry is exploited in regular regions, while sharp or ambiguous regions are not forced through the graph approximation. Third, the response quantities are coupled: the same signed field supplies gap, normal, and tangent frame. Fourth, the cost is output sensitive: expensive curved-patch work is performed only for near candidates and SDF sampling, while an online dense response query is constant time per active contact. Fifth, failures are contained: a failed graph certificate, uncertain Newton iteration, or invalid SDF cell only selects a more general local query.

3.2. Generic contact-element representation

The backend represents every possible moving contact feature through a contact element interface. A contact element B is a rigidly transformed geometric object with pose $q = (x, R)$, a bounding radius or swept bound for broad-phase culling, a local-patch provider, and, when available, a surface-velocity provider. The local-patch query maps a pose and a nearby spatial region to one or more curved primitives

$$\mathcal{E}_B(q, \mathcal{O}) = \{\mathcal{P}_j(q) : X_j^q(U_j) \cap \mathcal{O} \neq \emptyset\}, \qquad X_j^q(u) = RX_j(u) + x. \tag{2}$$

The same abstraction covers spheres, rollers, box-like rigid meshes, local CAD patches, and analytic surfaces. Scene-specific models only implement the provider; the detector and response loop operate on the returned primitives.

Each primitive is lifted into a curved jet

$$\mathcal{P}_i = \left(X_i, N_i, \kappa_i^{\max}, \varepsilon_i^x, \varepsilon_i^N, \mathcal{C}_i^N\right), \tag{3}$$

where $X_i : U_i \to \mathbb{R}^3$ is a local curved patch over a reference domain U_i, N_i is a smooth or piecewise smooth normal field, $\kappa_i^{\max}$ is a curvature bound, ε_i^x and ε_i^N are position and normal error bounds relative to the intended geometry, and $\mathcal{C}_i^N$ is a normal cone enclosing $N_i(U_i)$. Analytic and CAD-like primitives provide these quantities directly. For triangulated input, the implementation can use either a cubic Bezier triangle or a quadratic normal-height primitive. In the Bezier case,

$$X_i(u, v, w) = \sum_{a+b+c=d} B_{abc}^d(u, v, w) c_{abc}, \qquad d = 3, \qquad u + v + w = 1, \tag{4}$$

where B_{abc}^d are Bernstein basis functions. Vertex control points coincide with the triangle vertices, while edge and interior control points are chosen from vertex normals or a local least-squares fit. This construction is related to curved PN triangles [42]. The normal field is either computed from the patch,

$$N_i = \frac{X_{i,u} \times X_{i,v}}{\|X_{i,u} \times X_{i,v}\|}, \tag{5}$$

or represented by a separate fitted normal field with consistency checks.

The approximation assumption used for the convergence statement is deliberately separated from the robustness logic.

Assumption 1 (Curved-jet approximation)**.** *Let S be locally C^3 and sampled with characteristic spacing h. The curved jet primitive approximates S with*

$$\|X_i - X\|_{L^\infty(U_i)} \leq C_x h^3, \qquad \angle(N_i, N) \leq C_N h^2, \tag{6}$$

away from intentionally sharp features. In implementation, the constants are replaced by computable conservative bounds ε_i^x and ε_i^N.

Assumption 1 is used only to state the high-order behavior on regular smooth patches. If a reliable smooth estimate is unavailable, the primitive can be downgraded to the original triangle with larger error bounds, or the pair can be routed directly to the fallback hierarchy.

The global candidate search remains three dimensional. Each primitive receives an inflated bounding box

$$\mathrm{AABB}_i^{\mathrm{inf}} = \mathrm{AABB}(X_i(U_i)) \oplus \left(\varepsilon_i^x + \hat{d} + r_i + \varepsilon_i^{\mathrm{motion}}\right), \tag{7}$$

where $\hat{d}$ is the detector activation distance, r_i is an optional physical shell or offset radius, and $\varepsilon_i^{\text{motion}}$ is a swept bound over the time step. BVH traversal produces candidate pairs. A linear triangle distance is used only as a conservative first rejection test:

$$d_{\text{lin}}(T_i, T_j) > \hat{d} + \varepsilon_i^x + \varepsilon_j^x + r_i + r_j \quad \Longrightarrow \quad \text{reject.} \tag{8}$$

Pairs that remain enter the contact-aligned narrow phase.

3.3. Contact-aligned local graph detector

The narrow phase first builds the contact-aligned frame in which the graph reduction is attempted. For a candidate pair $(\mathcal{P}_A, \mathcal{P}_B)$, preliminary closest points (x_A^0, x_B^0) are computed from the linear triangles or a low-cost curved query, and

$$x_c = \frac{1}{2}(x_A^0 + x_B^0) \tag{9}$$

defines the frame origin. If $\|x_B^0 - x_A^0\|$ is sufficiently large, the contact direction is

$$n_c = \frac{x_B^0 - x_A^0}{\|x_B^0 - x_A^0\|}. \tag{10}$$

If the preliminary gap is near zero, representative opposing normals are used,

$$n_c = \frac{\bar{N}_A - \bar{N}_B}{\|\bar{N}_A - \bar{N}_B\|}. \tag{11}$$

The local normal convention is fixed throughout the derivation: N_A points from body A toward the gap and $-N_B$ points from body B toward the same gap. If stored input normals do not satisfy this convention for a candidate pair, they are flipped locally before graphability, normal averaging, and response-frame construction are evaluated. A deterministic rule chooses an orthonormal basis (t_1, t_2, n_c).

The associated contact-plane coordinates are

$$\xi(x) = \begin{bmatrix} t_1^T(x - x_c) \\ t_2^T(x - x_c) \end{bmatrix}, \qquad h(x) = n_c^T(x - x_c). \tag{12}$$

This plane is not a global surface parameterization. It is created for one local candidate pair and is used only while its validity certificate holds.

The certificate asks whether each surface can be written as a height graph over this plane. Let $\pi_c(x) = \xi(x)$ be the projection onto the contact plane. A patch is graphable if $\pi_c \circ X$ is nonsingular and single valued on the clipped local patch. A sufficient differential condition is that the patch normal never becomes tangent to the projection direction.

Definition 1 (Normal-cone graphability). *A pair $(\mathcal{P}_A, \mathcal{P}_B)$ is μ-graphable in the contact frame (t_1, t_2, n_c) if*

$$\inf_{p \in U_A} N_A(p) \cdot n_c \geq \mu > 0, \qquad \inf_{q \in U_B} -N_B(q) \cdot n_c \geq \mu > 0. \tag{13}$$

The two inequalities use the same local normal convention, so the scalar graph gap in Eq. (18) and the smooth normal in Eq. (21) do not depend on arbitrary mesh-normal storage choices. In practice, the condition is evaluated by normal cones. If the normal cone of patch A has axis $\bar{N}_A$ and angular radius θ_A, then

$$\angle(\bar{N}_A, n_c) + \theta_A < \frac{\pi}{2} - \delta \tag{14}$$

ensures $N_A \cdot n_c \geq \mu = \sin\delta$. A similar test is applied to $-N_B$.

Theorem 1 (Certified local graph reduction)**.** *Let $X : U \subset \mathbb{R}^2 \to \mathbb{R}^3$ be a C^1 regular patch with unit normal N. If $N(p) \cdot n_c \geq \mu > 0$ for all $p \in U$, then the orthogonal projection $\pi_c \circ X$ is locally invertible everywhere on U. Moreover, the inverse graph map has condition number bounded by*

$$\|D(\pi_c \circ X)^{-1}\| \leq \frac{C_X}{\mu}, \tag{15}$$

where C_X depends only on the local parameter scaling of X. If the clipped patch has no self-overlap under π_c, then X can be written as a single height graph $X(\xi) = \xi + h(\xi)n_c$ over its projected domain. If both contacting patches satisfy this condition over a common contact plane and have overlapping projections, their regular two-patch gap is represented by the scalar field $g(\xi) = h_B(\xi) - h_A(\xi)$ on the overlap.

Sketch. The Jacobian of $\pi_c \circ X$ loses rank exactly when a nonzero tangent vector of the surface projects to zero, i.e., when that tangent vector is parallel to n_c. This occurs only if n_c lies in the tangent plane, equivalently $N \cdot n_c = 0$. The lower bound $N \cdot n_c \geq \mu$ therefore prevents singular projection. The inverse function theorem gives local graphability and an inverse norm controlled by the angle between n_c and the tangent plane. The no-overlap condition upgrades the local charts to a single-valued graph over the clipped projected domain. Applying the construction to both patches over a shared contact plane yields the common gap field. □

When the certificate holds, the detector evaluates a two-dimensional graph gap. Let $D_A = \pi_c(X_A(U_A))$ and $D_B = \pi_c(X_B(U_B))$ and define the projected interaction domain

$$D_{AB} = D_A \cap D_B. \tag{16}$$

For a point $\xi \in D_{AB}$, let $p_A(\xi)$ and $p_B(\xi)$ be the inverse projected coordinates on the two patches. The height functions are

$$h_A(\xi) = n_c^T(X_A(p_A(\xi)) - x_c), \qquad h_B(\xi) = n_c^T(X_B(p_B(\xi)) - x_c), \tag{17}$$

and the signed graph gap is

$$g_{AB}(\xi) = h_B(\xi) - h_A(\xi) - r_A - r_B. \tag{18}$$

Contact is active or near-active where $g_{AB}(\xi) < \hat{d}$. For detector sampling and response-field tabulation, the local contact point is obtained from

$$\xi^* = \arg\min_{\xi \in D_{AB}} g_{AB}(\xi). \tag{19}$$

The current graph solver approximates D_{AB} by the overlap of projected reference triangles; a more exact version can clip projected Bezier edge curves or use adaptive sampling. Newton or safeguarded quasi-Newton iterations minimize g_{AB} inside the overlap, and active-set logic handles overlap edges. If the minimum is below $\hat{d}$, a detector sample is emitted:

$$c_s = (x_A, x_B, n, g, p_A, p_B, \text{type}, \text{certificate}), \tag{20}$$

where $x_A = X_A(p_A(\xi^*))$, $x_B = X_B(p_B(\xi^*))$, $g = g_{AB}(\xi^*)$, and n is either n_c or the normalized smooth opposing normal

$$n(\xi) = \frac{N_A(p_A(\xi)) - N_B(p_B(\xi))}{\|N_A(p_A(\xi)) - N_B(p_B(\xi))\|}. \tag{21}$$

3.4. Fallback and validity control

The graphability certificate is a fast path rather than an input requirement. If graphability fails or graph minimization is uncertain, the algorithm solves the generic curved patch-patch closest problem

$$\min_{p \in U_A,\, q \in U_B} \frac{1}{2}\|X_A(p) - X_B(q)\|^2. \tag{22}$$

Let $r = X_A(p) - X_B(q)$ and $J_A = DX_A(p)$, $J_B = DX_B(q)$. Interior stationary points satisfy

$$J_A^T r = 0, \qquad J_B^T r = 0. \tag{23}$$

A Gauss-Newton step solves

$$\begin{bmatrix} J_A^T J_A & -J_A^T J_B \\ -J_B^T J_A & J_B^T J_B \end{bmatrix} \begin{bmatrix} \Delta p \\ \Delta q \end{bmatrix} = - \begin{bmatrix} J_A^T r \\ -J_B^T r \end{bmatrix}. \tag{24}$$

The parameters are projected back to the reference domains. If a minimizer lies on the boundary of one patch, the local active set is reduced to a curve-surface problem; if it lies on both boundaries, it becomes a curve-curve problem. Vertex cases are handled as further active-set degeneracies. Open surface boundaries and sharp features therefore need not be globally pre-classified for validity; they appear naturally through the active set. Feature-aware normal splitting can still be used to prevent artificial smoothing across known sharp creases.

The final validity layer prevents uncertain numerical results from being silently accepted. For Bezier primitives, the convex hull property gives conservative bounds on spatial extent. Interval ranges or Bernstein bounds on the gap can reject subdomains. Interval Newton or local subdivision is invoked when the standard Newton residual remains large, when the Hessian is ill-conditioned, or when the gap interval straddles the activation threshold. The final emergency fallback is the original linear triangle distance or CCD query with inflated tolerances. The resulting hierarchy is

$$\begin{gathered} \text{BVH} \to \text{linear reject} \to \text{graph certificate} \to \text{2D graph solve} \\ \to \text{4D patch solve} \to \text{certified subdivision}. \end{gathered} \tag{25}$$

Open and closed surfaces are treated uniformly at the search level. Closed surfaces may provide a globally consistent outward normal, but this is not required for building the contact frame. Open surfaces do not have a global signed distance, yet the local gap Eq. (18) remains well defined whenever graphability holds. The configuration-space response field is built only on a chosen valid band in a local $SE(3)$ chart from certified local gap evaluations; it is not assumed to be a global volumetric distance field for the entire object. If no sharp-feature information is available, large normal cones naturally reduce graphability and send the pair to the generic fallback.

3.5. Configuration-space response field

The rigid-body backend adds a signed-gap response field after the local detector. The general response coordinate is a local chart of $SE(3)$, written as $q = (x, \theta) \in \mathbb{R}^6$, where x is translation and θ is a rotation-vector coordinate around a reference orientation. The configuration-space field in Eq. (1) can be tabulated on a narrow band Ω_Δ of this chart. For a general orientation-dependent contact element, tensor-product cubic interpolation over the six local coordinates gives

$$\tilde{\phi}(q) = \mathcal{I}_6[\phi_{\mathbf{i}}](q), \qquad \nabla_q \tilde{\phi}(q) = \left[\nabla_x \tilde{\phi}(q)^T,\, \nabla_\theta \tilde{\phi}(q)^T\right]^T. \tag{26}$$

The contact normal used by the rigid friction law is obtained from the translational gradient,

$$\tilde{n}(q) = \frac{\nabla_x \tilde{\phi}(q)}{\|\nabla_x \tilde{\phi}(q)\|}. \tag{27}$$

The rotational gradient is retained as part of the general response coordinate because an orientation-dependent contact element can use it for generalized force or torque assembly. For a tensor-product cubic response cell with normalized local coordinates $s = (s_1, \ldots, s_d) \in [0,1]^d$, the interpolant has the compact multi-index form

$$\tilde{\phi}(q) = \sum_{\alpha \in \{0,1,2,3\}^d} a_\alpha \prod_{\ell=1}^{d} s_\ell^{\alpha_\ell}, \qquad d \le 6, \tag{28}$$

where the coefficients a_α are obtained from the local grid stencil in the chosen response chart. The interpolated value gives the signed gap used by the rigid contact response, and the normalized translational gradient gives the force normal and tangential friction frame. If a query leaves the valid interpolation band, crosses a cell whose detector samples are uncertified, or satisfies $\|\nabla_x \tilde{\phi}\| < \eta$ for a prescribed tolerance η, the backend falls back to the direct CALG detector query.

This response layer separates detection thickness from physical force activation. The detector distance $\hat{d}$ is used only to collect near candidates and to build a sufficiently wide interpolation band. The normal force in the rigid response is activated by physical penetration,

$$p(q) = \max(0, -\tilde{\phi}(q)), \tag{29}$$

with a small release tolerance used only for contact-state persistence.

For an active rigid contact element, the dynamic backend can query a small local surface-contact response patch rather than a single representative point. Let $\mathcal{Q}_c = \{(w_i, \delta x_i)\}_{i=1}^{Q}$

denote fixed local quadrature offsets in the contact tangent plane, with $w_i \geq 0$ and $\sum_i w_i = 1$. Each offset defines a nearby response configuration $q_i = (x + \delta x_i, R)$ and returns

$$\tilde{\phi}_i = \tilde{\phi}(q_i), \qquad \tilde{n}_i = \tilde{n}(q_i). \tag{30}$$

The rigid contact response is then assembled as a normalized response quadrature,

$$F_c = \sum_{i=1}^{Q} w_i \, \mathcal{F}(\tilde{\phi}_i, \tilde{n}_i, v_i, \mathcal{H}_i), \qquad \tau_c = \sum_{i=1}^{Q} (r_i - x) \times w_i \, \mathcal{F}(\tilde{\phi}_i, \tilde{n}_i, v_i, \mathcal{H}_i), \tag{31}$$

where r_i is the response point, v_i is the relative velocity, $\mathcal{H}_i$ is the local friction history, and $\mathcal{F}$ is the regularized Coulomb normal–tangential response used in the rigid update. The weights are normalized because this dynamic backend uses the local patch as a contact-element response object. Full area-measure integration, pressure centers, and average normals are verified separately by the static patch-contact equivalence check in Sec. 4. The dense response field introduces a grid-spacing interpolation error and a one-time preprocessing cost, but an online query is constant time per response sample. This preprocessing cost is excluded from the solver-loop timings reported in Sec. 4.

Proposition 1 (Cubic signed-gap consistency)**.** *Assume the exact configuration-space signed-gap field ϕ is C^4 on a local q-grid cell and $\|\nabla_x \phi\| \geq m > 0$ on that cell. Let $\tilde{\phi} = \mathcal{I}_d[\phi_{\mathbf{i}}]$ be a consistent tensor-product cubic interpolant on the chosen local response chart with $d \leq 6$. If the maximum grid spacing is Δ, then, for q inside the cell,*

$$|\tilde{\phi}(q) - \phi(q)| \leq C_0 \Delta^4, \qquad \|\nabla_x \tilde{\phi}(q) - \nabla_x \phi(q)\| \leq C_1 \Delta^3, \tag{32}$$

where C_0 and C_1 depend on fourth derivatives of ϕ and on the interpolation stencil. Consequently the normal error satisfies

$$\angle(\tilde{n}, n) \leq \frac{C_1}{m} \Delta^3 + O(\Delta^6), \qquad n = \frac{\nabla_x \phi}{\|\nabla_x \phi\|}. \tag{33}$$

Sketch. Tensor-product cubic interpolation gives the standard one-dimensional fourth-order value estimate in each coordinate direction, yielding the cellwise $O(\Delta^4)$ value bound and the differentiated $O(\Delta^3)$ bound for the translational gradient. The normal is the composition of $\nabla_x \tilde{\phi}$ with the smooth normalization map $v \mapsto v/\|v\|$ on the set $\|v\| \geq m$, whose Lipschitz constant is bounded by $1/m$ up to higher-order terms. □

3.6. Error estimates, algorithms, and complexity

The error model separates detector geometry from response-field interpolation. Let g be the true local signed surface gap and $\hat{g}_{\mathrm{CALG}}$ be the gap computed from curved jets before response interpolation. In a regular contact region with bounded curvature, if both position approximations satisfy $\|\hat{X}_i - X_i\| \leq \varepsilon_i^x$ and normal errors satisfy $\angle(\hat{N}_i, N_i) \leq \varepsilon_i^N$, then

$$|\hat{g}_{\mathrm{CALG}} - g| \leq \varepsilon_A^x + \varepsilon_B^x + Cd(\varepsilon_A^N + \varepsilon_B^N) + C_\kappa \left((\varepsilon_A^x)^2 + (\varepsilon_B^x)^2 \right), \tag{34}$$

Algorithm 1 Preprocessing for CALG–SDF

Require: Surface provider A, rigid contact element provider B, activation distance $\hat{d}$, response band $\Omega_\Delta \subset \mathrm{se}(3)$ if a rigid response field is requested
Ensure: Contact objects with curved primitives, inflated BVHs, and optional configuration-space signed-gap field

```
for each primitive T_i returned by a provider do
    Fit or import curved jet primitive P_i = (X_i, N_i, κ_i^max, ε_i^x, ε_i^N, C_i^N)
    Compute or bound AABB(X_i(U_i))
    Inflate using Eq. (7)
end for
Build a 3D BVH over inflated primitive boxes
if a rigid response field is requested then
    for each grid node q_i ∈ Ω_Δ do
        Evaluate φ(q_i) = g_CALG(S_A, q_i B) using graph, curved-patch, or certified fallback query
    end for
    Store tensor-product cubic interpolation data and validity/certificate flags
end if
Return providers, primitives, BVHs, response data if present, and feature tags
```

where d is the local separation scale and C_κ depends on curvature bounds. Near contact, $d \approx 0$, so the dominant detector term is the position approximation error. Under Assumption 1, the expected detector errors are

$$|\hat{g}_{\mathrm{CALG}} - g| = O(h^3), \qquad \angle(\hat{n}_{\mathrm{CALG}}, n) = O(h^2), \tag{35}$$

for regular smooth cells.

Combining the detector estimate with Proposition 1 gives the response-level estimate

$$|\tilde{\phi} - \phi| \le C_h h^3 + C_\Delta \Delta^4 + \varepsilon_{\mathrm{tab}}, \qquad \angle(\tilde{n}, n) \le C_N h^2 + C'_\Delta \Delta^3 + \varepsilon_{\nabla,\mathrm{tab}}, \tag{36}$$

where Δ is the response-field grid spacing and $\varepsilon_{\mathrm{tab}}$, $\varepsilon_{\nabla,\mathrm{tab}}$ collect numerical tabulation and finite-precision effects. Equation (36) is the accuracy statement for the CALG–SDF pipeline: the local graph detector controls geometric sampling error, while tensor-product cubic interpolation controls online response-field error. The estimate applies only on regular smooth cells where the curved-jet bounds are valid, the graphability certificate or certified fallback supplies reliable samples, and the response query remains inside its valid interpolation band. Sharp features, open-boundary active sets, and emergency fallback cells are covered by the conservative fallback logic rather than by this high-order estimate.

Let N be the number of input primitives, K_{cand} the number of conservative BVH candidate pairs, K_{graph} the number passing graphability, K_{fail} the number requiring generic or certified fallback, K_{act} the number of active or near-active rigid contacts, Q the fixed number of response samples per active contact element, and G the number of response-field grid nodes if a field is built. The one-time broad-phase construction cost is

$$T_{\mathrm{BVH,build}} = O(N \log N). \tag{37}$$

When moving geometry changes primitive bounds, a timestep may additionally require a BVH refit or rebuild cost $T_{\mathrm{BVH,update}}$; for the fixed-surface rigid validation scenes this term is a refit or zero-cost lookup rather than a full rebuild. The online query cost per step is

$$T_{\mathrm{step}} = T_{\mathrm{BVH,update}} + K_{\mathrm{cand}} C_{\mathrm{cert}} + K_{\mathrm{graph}} C_{2\mathrm{D}} + K_{\mathrm{fail}} C_{\mathrm{fallback}} + K_{\mathrm{act}} Q C_{\mathrm{sdf}}, \tag{38}$$

where C_{cert} is the inexpensive normal-cone and linear-rejection cost, $C_{2\mathrm{D}}$ is the graph gap minimization cost, C_{fallback} is the cost of generic patch-patch and certified methods, and $C_{\mathrm{sdf}} = O(1)$ for a dense tensor-product cubic response query. In the implemented rigid validation backend, Q is a small constant and the quadrature weights are normalized as in Eq. (31). The one-time response-field construction cost is

$$T_{\mathrm{sdf,build}} = O(G\, C_{\mathrm{sample}}), \tag{39}$$

where C_{sample} is the average cost of a certified CALG sample at a grid node. These preprocessing and memory costs are reported separately and are not included in the solver-loop timing. The method is advantageous when regular smooth patch contact dominates the detector workload and when repeated rigid response queries amortize the response-field construction cost.

4. Numerical results

This section reports the executable evidence for the proposed rigid frictional contact-geometry backend. The numerical sequence follows the logic of the paper. First, a static patch-contact equivalence check verifies that the local graph representation recovers area-level contact quantities over an active patch. Second, analytic inclined-plane controls verify that the friction update reproduces Coulomb sticking, sliding, and threshold behavior. Third, four engineering scenes compare CALG–SDF directly with Linear-triangle BVH and PN-triangle BVH under the same rigid-body update, so that any force or velocity jump is attributable to the contact-geometry module. Fourth, three selected engineering scenes are reproduced against Chrono and Recurdyn to evaluate trajectory-level agreement in displacement and velocity trends with independent dynamics solvers. Finally, native C++ timings quantify the solver-loop cost of the implemented CALG–SDF backend.

4.1. Static patch-contact equivalence check

The first numerical check isolates the geometric part of the method before introducing frictional dynamics. Two graphable but geometrically complex analytic patches are constructed over a shared contact plane. Their signed graph gap is prescribed so that

$$\Omega_c = \{\xi : g_{AB}(\xi) \le 0\} \tag{40}$$

is an ellipse in the local contact coordinates. Inside this active patch, the maximum compression is 4.0×10^{-4} m and the gap varies smoothly to zero at the boundary. The comparison uses the contact-plane measure $d\xi$ and the regularized pressure

$$p_n(\xi) = k_n \max(0, -g_{AB}(\xi)), \qquad k_n = 2.0 \times 10^6\ \mathrm{N/m^3}. \tag{41}$$

The quantities being checked are therefore patch quantities rather than a single closest point: active area, $\int_{\Omega_c} g_{AB}(\xi)\, d\xi$, area-averaged gap, pressure resultant, pressure-weighted center, and pressure-weighted average normal.

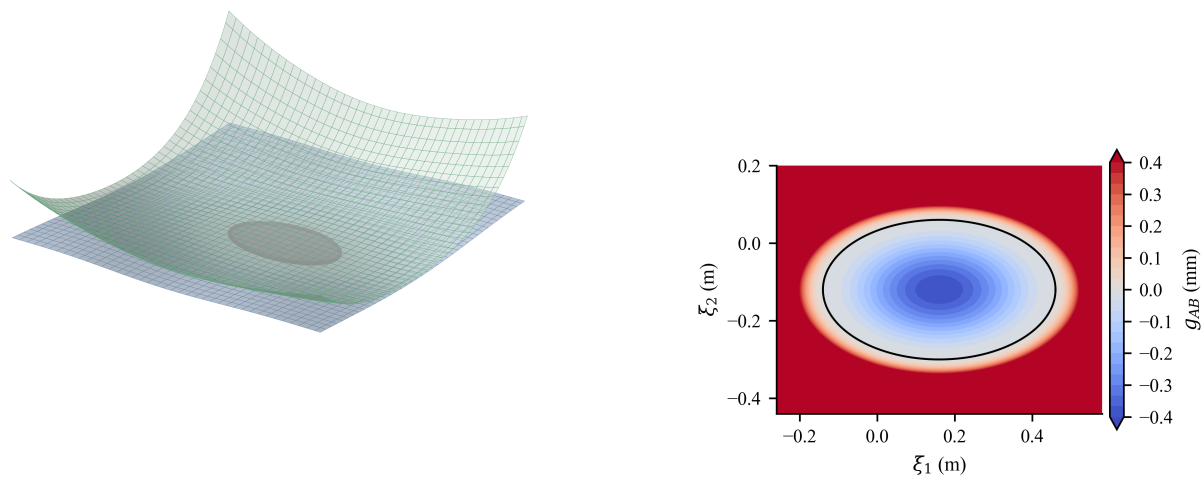

Figure 1: Static patch-contact equivalence setup. The active contact patch is an ellipse embedded in two graphable, geometrically complex analytic surfaces. The right panel shows the prescribed signed graph gap in the local contact coordinates; negative values are the active compressed region.

Table 2: Static patch-contact equivalence metrics. The reference active area, gap integral, mean gap, and normal force have analytic values; pressure center and average normal are compared with high-order quadrature.

Quantity	Reference	CALG graph	Error
Active area (m^2)	1.696460×10^{-1}	1.696460×10^{-1}	2.78×10^{-17}
$\int_{\Omega_c} g_{AB}\, d\xi$ (m^3)	-2.261947×10^{-5}	-2.261947×10^{-5}	3.05×10^{-20}
Mean gap (m)	-1.333333×10^{-4}	-1.333333×10^{-4}	1.63×10^{-19}
Normal force (N)	4.523893×10^{1}	4.523893×10^{1}	5.68×10^{-14}
Pressure center (m)	high-order quadrature	graph quadrature	1.02×10^{-15}
Mean normal angle (deg)	high-order quadrature	graph quadrature	$< 10^{-12}$

Table 2 shows that the CALG graph-region quadrature reproduces the analytic patch quantities to roundoff accuracy in this controlled case. The active-band graphability certificate remains well conditioned, with minimum $N_A \cdot n_c = 0.973$ and minimum $(-N_B) \cdot n_c = 0.973$ over the sampled contact band. This check does not replace dynamic validation; it establishes that the local graph representation can be used as an area-contact object when the active region is integrated rather than collapsed to a representative point.

Algorithm 2 Pairwise contact query and response-field refinement

Require: Candidate primitive pair $(\mathcal{P}_A, \mathcal{P}_B)$, rigid response configuration q, and optional response field

Ensure: Detector sample and response sample set $\mathcal{R}$

if linear distance rejects pair with conservative error bounds **then**
    return empty set
end if
Construct contact frame (t_1, t_2, n_c, x_c) using Eq. (12)
if normal-cone graphability test in Eq. (13) passes **then**
    Solve the two-dimensional graph gap problem in Eq. (18)
    if the solution is certified **then**
        $c_s \leftarrow$ direct CALG detector sample
    end if
end if
if no certified graph sample is available **then**
    Solve generic curved patch-patch problem Eq. (22)
    if patch solve is uncertain **then**
        Invoke interval, Bernstein, or subdivision fallback
    end if
end if
Let $(g_{\mathrm{CALG}}, n_{\mathrm{CALG}}, r_{\mathrm{CALG}})$ be the signed gap, normal, and response point from the certified detector path
Build a local response seed set $\mathcal{S} = \{(w_i, q_i, r_i)\}_{i=1}^{Q}$ around r_{CALG}; use $Q = 1$ for a representative-point response
$\mathcal{R} \leftarrow \emptyset$
for each $(w_i, q_i, r_i) \in \mathcal{S}$ **do**
    if configuration-space response field is active and valid at q_i **then**
        Query $\tilde{\phi}(q_i)$ and $\nabla_x \tilde{\phi}(q_i)$ by Eq. (26)
        Append $(w_i, \tilde{\phi}(q_i), \tilde{n}(q_i), r_i)$ to $\mathcal{R}$
    else
        Append the direct detector response $(w_i, g_{\mathrm{CALG}}, n_{\mathrm{CALG}}, r_i)$ to $\mathcal{R}$
    end if
end for
return detector sample and response sample set $\mathcal{R}$, or empty set

Algorithm 3 Global contact detection

Require: Contact objects A, B and rigid pose q

Ensure: Contact set $\mathcal{K}$

```
Refit or rebuild BVHs if geometry changed
𝒫 ← BVH candidate pairs between A and B(q)
𝒦 ← ∅
for each (𝒫_A, 𝒫_B) ∈ 𝒫 do
    𝒦 ← 𝒦 ∪ PairwiseContact(𝒫_A, 𝒫_B, q)
end for
Merge duplicate samples generated by neighboring primitives
Cluster neighboring detector samples that refer to the same physical contact region
return 𝒦
```

4.2. Analytic inclined-plane friction controls

Before using frictional dynamics to evaluate contact geometry, we first isolate the friction law on a planar inclined surface where the exact Coulomb solution is available. Positive displacement is measured downslope. For a block of mass m on a plane with angle θ, the normal load is $N = mg\cos\theta$ and the downslope gravitational force is $mg\sin\theta$. A block released from rest remains stuck if

$$mg\sin\theta \leq \mu_s mg\cos\theta, \tag{42}$$

and otherwise enters kinetic sliding with acceleration

$$a = g(\sin\theta - \mu_k\cos\theta). \tag{43}$$

These controls do not test geometric accuracy because the plane normal is exact for all contact modules. They test whether the implemented rigid-friction response correctly switches between static and kinetic regimes before the more demanding curved-surface comparisons.

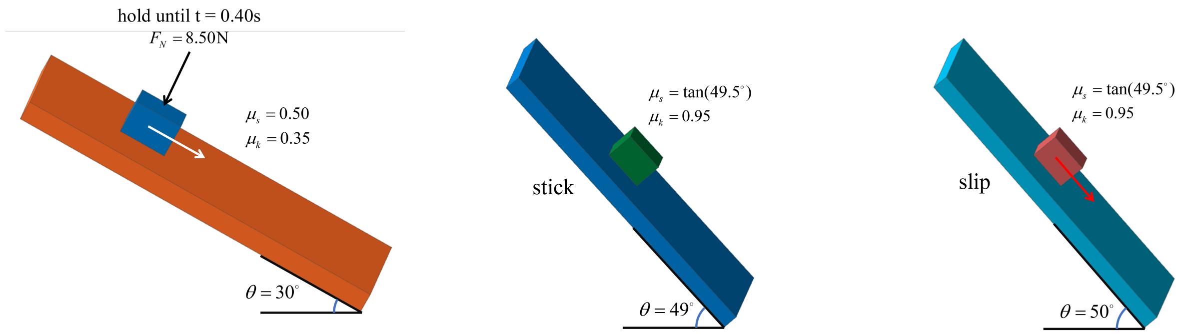

Figure 2: Model configuration for the analytic inclined-plane controls. The three views show the 30° release case and the two static-friction threshold cases with identical block–plane geometry. The threshold cases change the slope angle from 49° to 50° while keeping $\mu_s = \tan(49.5°)$.

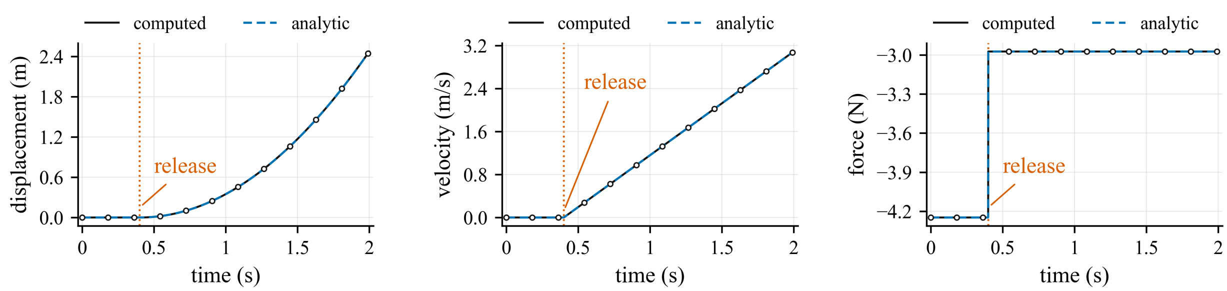

Figure 3: Release control on a 30° inclined plane. The block is held until $t = 0.40$ s and then released; because the static capacity is exceeded, it switches to kinetic sliding and follows Eq. (43). The black computed curve and blue dashed analytic curve are plotted separately, with markers on the computed samples to keep the coincident curves distinguishable.

Table 3: Analytic inclined-plane friction controls. Static margin is $\mu_s mg\cos\theta - mg\sin\theta$; positive values indicate sticking is admissible.

Case	θ (deg)	(μ_s, μ_k)	Margin (N)	Outcome	a (m/s^2)	$s(T)$ (m)
Release	30.0	(0.500, 0.350)	-0.657	slip	1.932	2.472
Threshold 49 deg	49.0	(1.171, 0.950)	0.132	stick	0	0
Threshold 50 deg	50.0	(1.171, 0.950)	-0.132	slip	1.524	0.762

4.3. Engineering scenes and compared geometry modules

The dynamic backend comparison isolates the contact-geometry module while keeping the rigid-body update fixed. Linear-triangle BVH uses the piecewise-linear triangle mesh and face normals. PN-triangle BVH evaluates a fixed-tessellation PN-triangle approximation constructed from the same input geometry. CALG–SDF uses the contact-aligned local graph detector, certified fallback hierarchy, and configuration-space signed-distance response layer to return the contact point, signed gap, contact normal, and tangential frame used by the same regularized Coulomb update.

The four scenes cover common rigid contact mechanisms in which friction is strongly coupled to contact-normal quality: rolling elements in raceways, cam–roller follower contact, ball or cylindrical heads in guide slots, and ball-and-socket pendulum contact. The contact point crosses multiple surface cells in each scene, so a facet-based contact-geometry module changes the normal and tangential basis as a function of mesh topology rather than only physical geometry. The reproducibility package stores the exact step counts, timestep, friction coefficient, contact radius, detector distance, stiffness, and mesh resolution for each generated artifact.

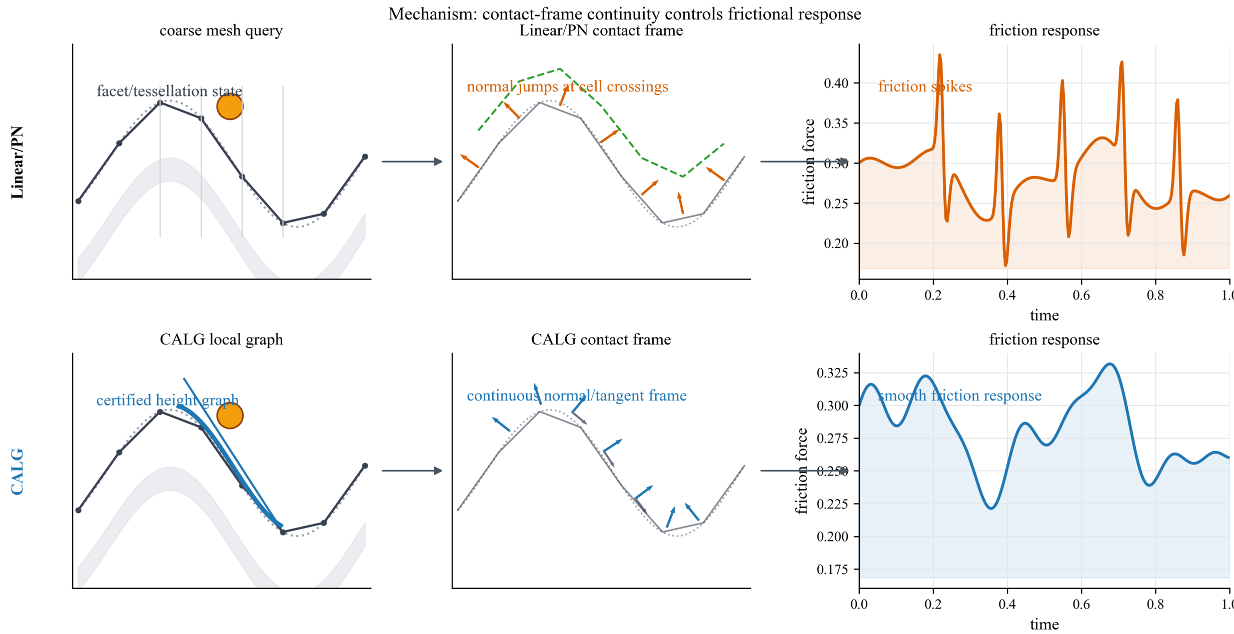

Figure 4: Mechanism-level comparison between facet-based contact-geometry modules and CALG–SDF. Linear-triangle BVH and fixed-tessellation PN-triangle BVH inherit cell-crossing changes in the contact normal or tangent frame, which can enter the regularized Coulomb response as force spikes. CALG instead certifies a contact-aligned local height graph and returns a smooth normal and tangent frame, producing a smoother friction response in the same rigid-body update.

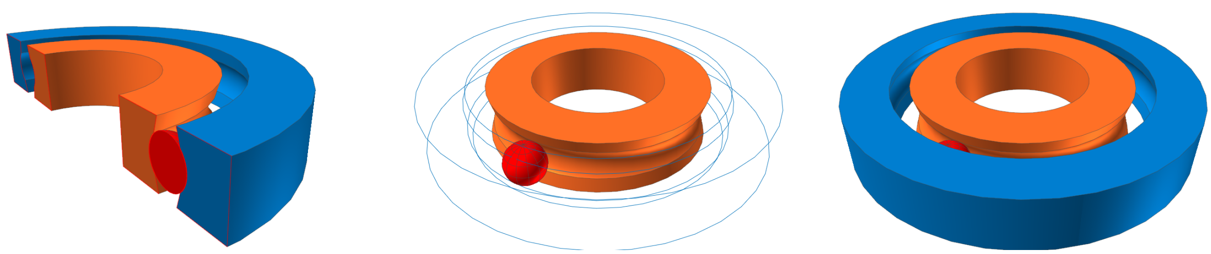

Figure 5: Bearing raceway model configuration used for the validation scene. The three views show the imported raceway geometry and rolling element configuration used by Recurdyn, Chrono, and CALG.

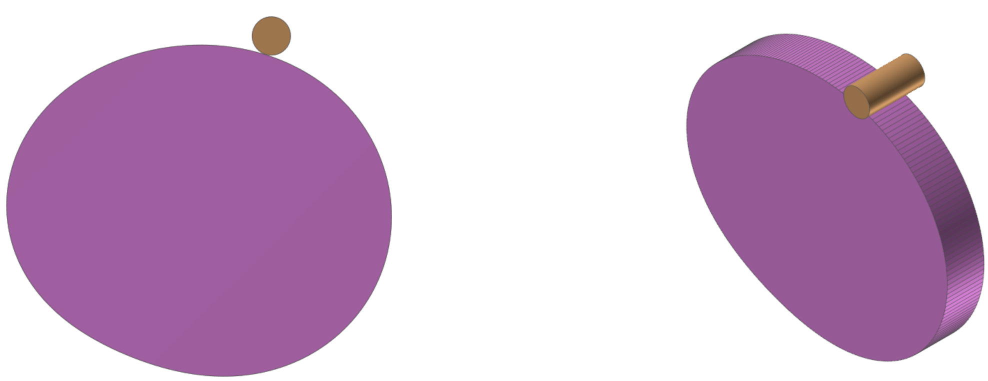

Figure 6: Cam–roller follower model configuration used for the contact-geometry comparison. The two views show the non-circular cam profile and roller follower geometry that produces sliding and rolling contact over a faceted collision representation.

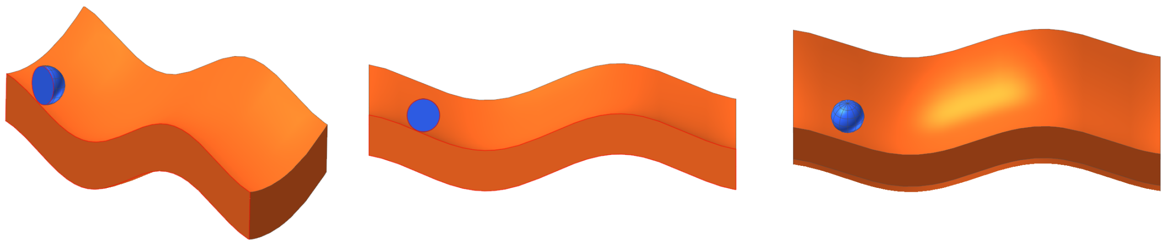

Figure 7: Guide slot model configuration used for the validation scene. The three views show the slot path, moving contact body, and contact geometry used by Recurdyn, Chrono, and CALG.

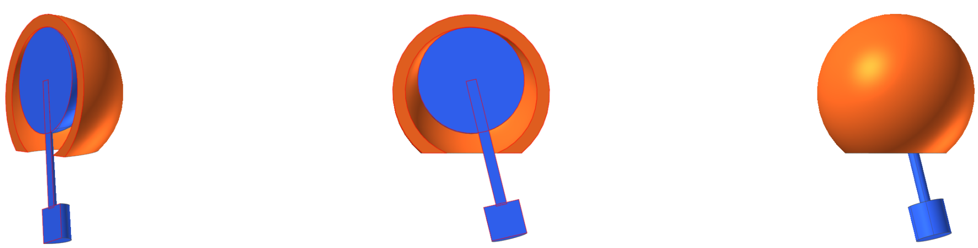

Figure 8: Ball-and-socket pendulum model configuration used for the validation case. The three views show the socket, moving contact ball, and attached pendulum body used by Recurdyn, Chrono, and CALG.

4.4. Dynamic friction comparison against Linear-triangle BVH and PN-triangle BVH

The main dynamic comparison runs the same rigid-body friction update with the three contact-geometry modules. All four dynamic backend comparison scenes are run for 0.8 s with 801 stored frames. The reported quantities are the maximum consecutive contact-normal angle jump, maximum normal-force jump, maximum friction-force jump, tangential-speed jitter, and stick–slip switch count. Table 4 reports ratios of the baseline value over the CALG value; values above one mean that the baseline produces a larger jump or jitter in the same scene. The stick–slip switch metric is a count ratio and is interpreted only as a discrete chatter indicator, not as a continuous error norm.

Figures 9–12 plot the independent contact-geometry comparison time histories for the four engineering scenes. Each figure reports the normal-angle jump, friction-force norm, and tangential-speed jitter from the same locked dynamic artifact used to compute Table 4. In these four figures, the black solid curve denotes CALG, the green dashed curve denotes PN-triangle BVH, and the orange dash-dot curve denotes Linear-triangle BVH. The per-scene curves make the mechanism visible: CALG keeps the contact-normal angle and tangential-speed jitter smooth, whereas the faceted and fixed-tessellation baselines show repeated jumps that coincide with elevated friction-force variation.

Table 4: Dynamic rigid-friction contact-geometry module comparison. Ratios are baseline/CALG under identical body, timestep, penalty, and friction parameters.

Model	Metric	Linear-triangle BVH/CALG	PN-triangle BVH/CALG
Bearing raceway	Δn	2.20×10^3	826
	ΔF_n	1.50×10^3	636
	ΔF_t	50.7	62.6
	v_t jitter	2.20×10^3	301
	stick–slip switch ratio	1	1
Cam–roller follower	Δn	71.9	40.2
	ΔF_n	44.2	24.6
	ΔF_t	37.7	35.8
	v_t jitter	23.8	10.5
	stick–slip switch ratio	322	0
Ball-and-socket pendulum	Δn	348	83.9
	ΔF_n	183	42.8
	ΔF_t	108	56.9
	v_t jitter	425	38.6
	stick–slip switch ratio	1	769
Guide slot	Δn	1.08×10^3	589
	ΔF_n	708	377
	ΔF_t	60.8	58.6
	v_t jitter	569	334
	stick–slip switch ratio	1	1

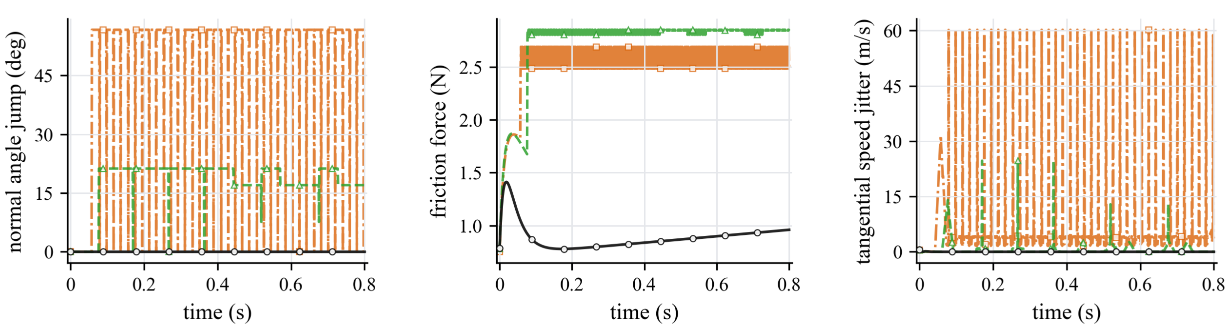

Figure 9: Independent contact-geometry comparison curves for the bearing raceway case. The body state, timestep, penalty stiffness, friction coefficient, and regularized Coulomb update are identical; only the contact-geometry module changes.

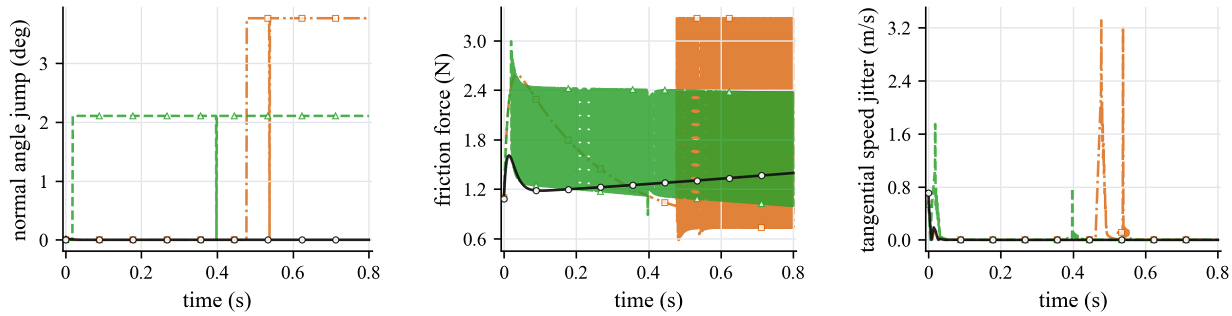

Figure 10: Independent contact-geometry comparison curves for the cam–roller follower case. Linear-triangle BVH and PN-triangle BVH produce larger normal and friction-force variations when the contact point traverses cam cells.

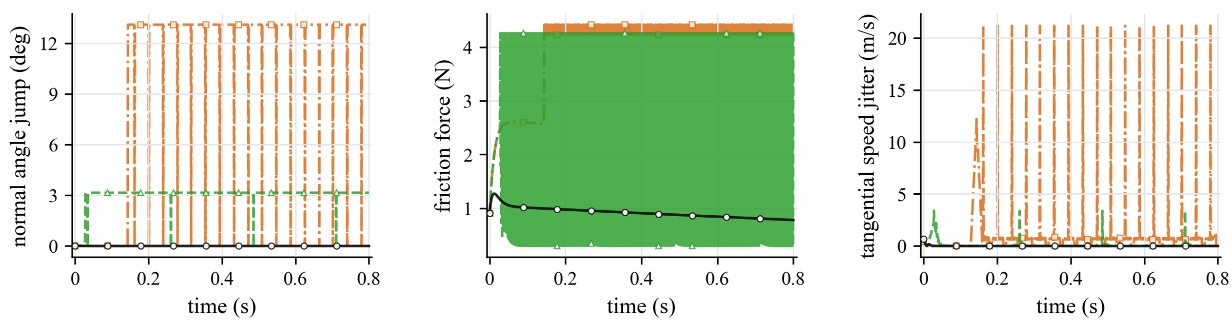

Figure 11: Independent contact-geometry comparison curves for the ball-and-socket pendulum case. The smooth CALG contact frame suppresses the high-frequency tangential-speed jitter and discrete friction-state changes observed in the fixed-tessellation baselines.

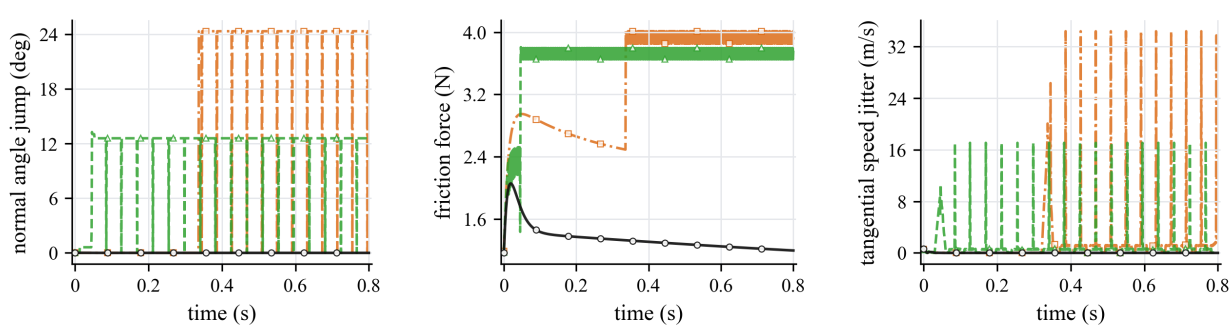

Figure 12: Independent contact-geometry comparison curves for the guide slot case. Cell-crossing normal changes in the Linear-triangle BVH and PN-triangle BVH modules enter directly into friction-force and tangential-speed jitter.

The dynamic results support the central mechanism of the method. Linear-triangle BVH consistently generates much larger geometric and force discontinuities: the normal-angle jump ratio ranges from 71.9 to 2.20×10^3, the normal-force jump ratio from 44.2 to 1.50×10^3, the friction-force jump ratio from 37.7 to 108, and the tangential-speed jitter ratio from 23.8 to 2.20×10^3. PN-triangle BVH improves over Linear-triangle BVH in several scenes, but it remains tied to a tessellated approximation and still produces 40.2–826 times larger normal-angle jumps and 35.8–62.6 times larger friction-force jumps than CALG. The large stick–slip switch ratios for Linear-triangle BVH in the cam–roller follower case and for PN-triangle BVH in the ball-and-socket pendulum case show that smoother geometry affects not only the gap value but also the discrete friction state.

4.5. External validation against Chrono and Recurdyn

The same-scene backend comparison isolates the geometric input to the friction response, but it does not by itself show that the resulting motion is consistent with independent dynamics engines. We therefore reproduced three selected engineering scenes in Chrono, Recurdyn, and the CALG detector-driven rigid contact model: the guide slot, the ball-and-socket pendulum, and the bearing raceway. Each case uses the same scene description, rigid-body mass and geometry, initial state, friction coefficient, and timestep where applicable. In the CALG C++ runs, the local curved mesh detector supplies candidate/contact work and a configuration-space SDF refines the signed gap and normal over a small surface-contact response patch; the regularized normal and friction responses are assembled by the normalized quadrature in Eq. (31). Chrono and Recurdyn use different contact envelopes and regularizations, so the validation target is trajectory-level agreement in displacement and velocity trends rather than pointwise equality of raw contact force.

In all trajectory-validation figures, the black solid curve denotes CALG, the blue dashed curve denotes Chrono, and the orange dash-dot curve denotes Recurdyn; sparse open markers are used only to separate nearly overlapping traces. The plotted directions are the three Cartesian ball coordinates, and displacements are relative to each solver's initial ball coordinate.

The guide slot case tests confined sliding under wall contact, so the expected response is sustained sliding along the slot while the side wall confines transverse motion. Figures 13–15 show that the three solvers produce the same sliding direction, comparable transverse confinement, and bounded acceleration after visualization smoothing. This visual agreement in the dominant displacement and velocity trends is the intended validation target for the selected slot model. The limitation is that acceleration is a diagnostic channel rather than a strict pointwise validation target: CALG, Chrono, and Recurdyn export or reconstruct acceleration through different contact-force and envelope definitions.

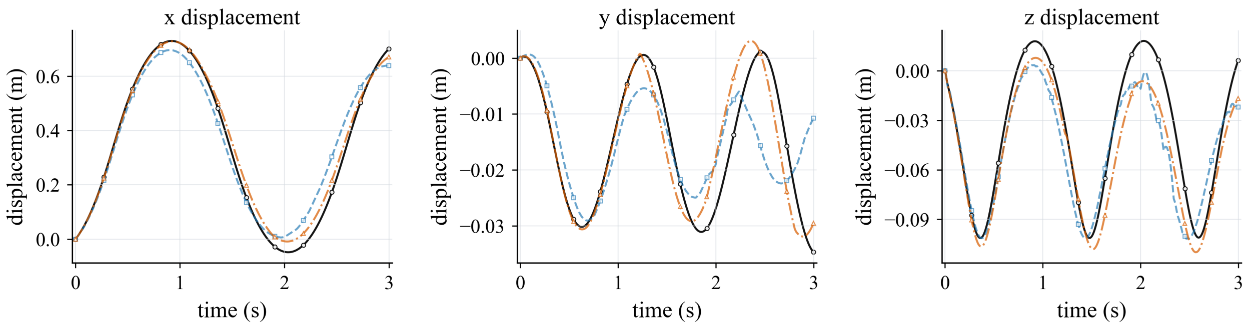

Figure 13: Guide slot validation over 3 s: ball displacement in the x, y, and z directions. The CALG result is shown as the black solid curve; displacements are relative to each solver's initial ball coordinate.

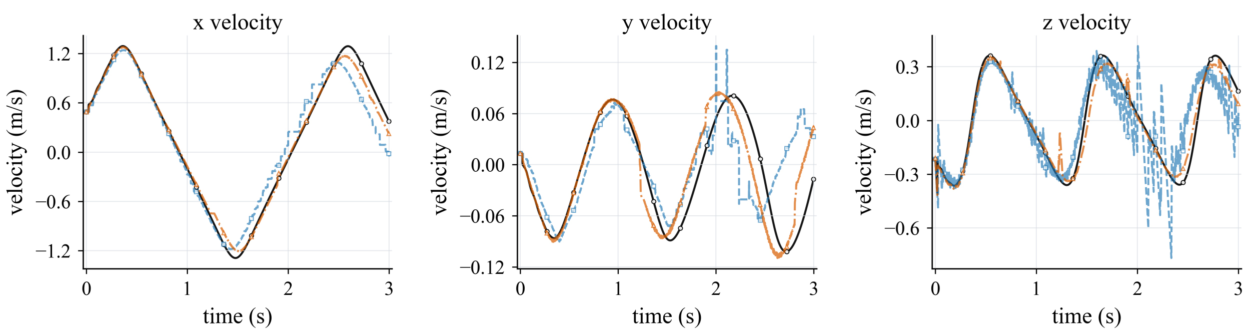

Figure 14: Guide slot validation over 3 s: ball velocity in the x, y, and z directions. The CALG result is shown as the black solid curve.

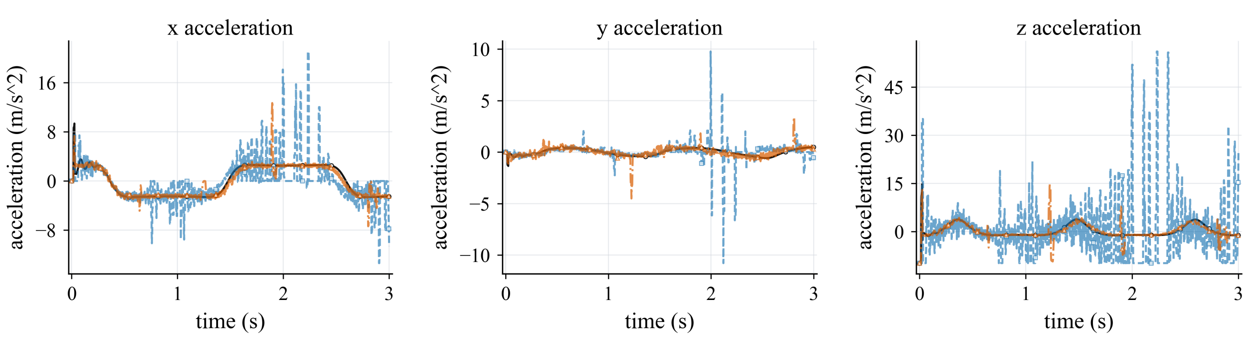

Figure 15: Guide slot validation over 3 s: ball acceleration in the x, y, and z directions. For CALG, acceleration is the effective per-step value exported by the C++ backend after the normal-constraint and gap-clamp update; for Chrono it is evaluated from the reported contact force and gravity; for Recurdyn it is taken from the exported body-acceleration channels. The plotted curves use a 21-frame centered average for visualization.

The ball-and-socket pendulum case tests gravity-driven swing under deep socket confinement, so the expected response is pendular motion with socket-limited contact. Figures 16–18 compare the same 5 s window after matching the Recurdyn moving-body mass distribution to the rod-plus-cylinder assembly used by Chrono and CALG. The first vertical pendulum peak occurs at 1.1438 s in Chrono, 1.1304 s in CALG, and 1.1252 s in Recurdyn, with peak vertical displacements of 0.1255, 0.1330, and 0.1383 m, respectively. These comparable peak timings and amplitudes support the physical reasonableness of the CALG trajectory. The remaining discrepancy is concentrated in re-contact transients and raw acceleration, where the three solvers use different contact regularizations and contact-envelope definitions.

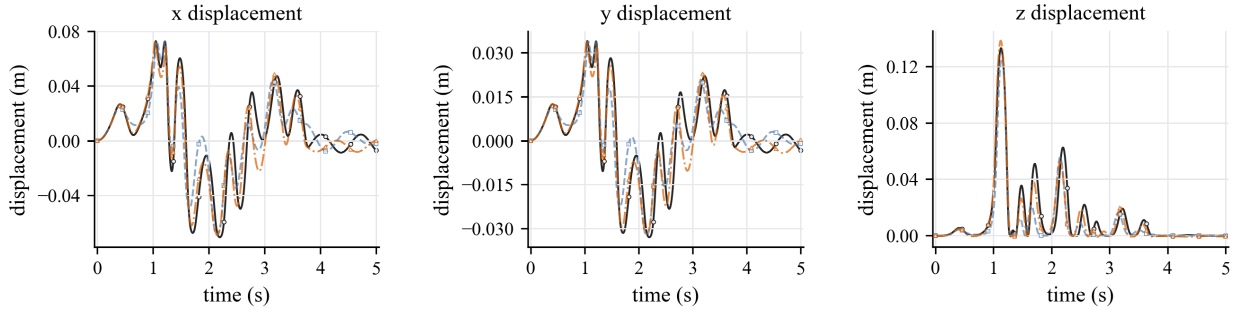

Figure 16: Ball-and-socket pendulum validation over 5 s: ball displacement in the x, y, and z directions. The Recurdyn model uses an equivalent distal payload with the same rod-plus-cylinder mass center and inertia as the Chrono and CALG composite body. The CALG result is shown as the black solid curve; displacements are relative to each solver's initial ball coordinate.

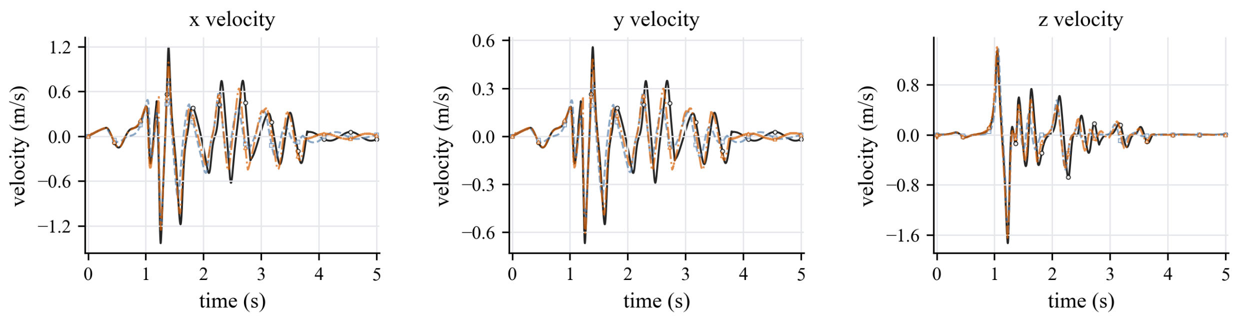

Figure 17: Ball-and-socket pendulum validation over 5 s: ball velocity in the x, y, and z directions. The dominant motion trend is consistent across Chrono, Recurdyn, and CALG, while re-contact transients remain solver dependent.

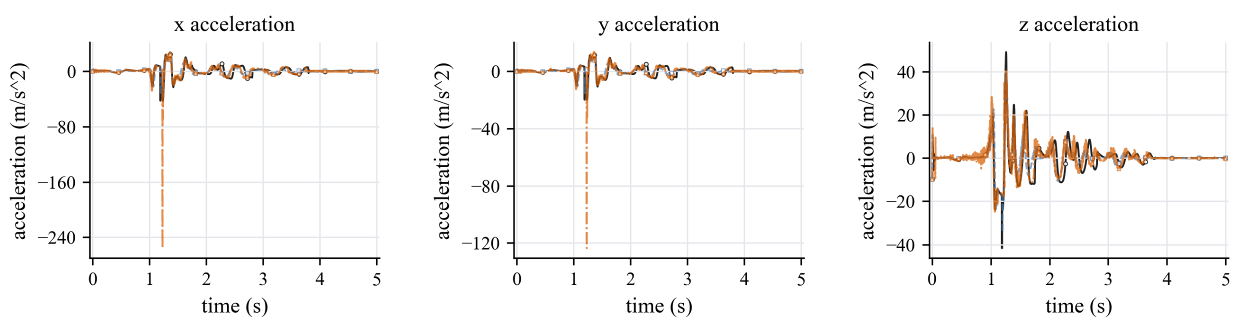

Figure 18: Ball-and-socket pendulum validation over 5 s: ball acceleration in the x, y, and z directions. Accelerations are shown as diagnostic channels for the contact transient; the displacement and velocity curves provide the primary trajectory-level validation because the three solvers use different contact regularizations and contact-envelope definitions.

The bearing raceway case tests raceway-driven in-plane rolling/sliding, with the axial z direction retained as a small diagnostic channel. Figures 19–21 show that the dominant x and y displacement and velocity curves follow the imposed raceway motion over 5 s in all three solvers. The small axial response is retained to expose contact-envelope differences rather than to define the main motion. The Recurdyn contact stiffness is raised to 1.0×10^8 with damping 1.5×10^4 to reduce vertical over-penetration, but raw contact force and post-processed gap remain envelope-dependent and are not used as strict pointwise validation targets.

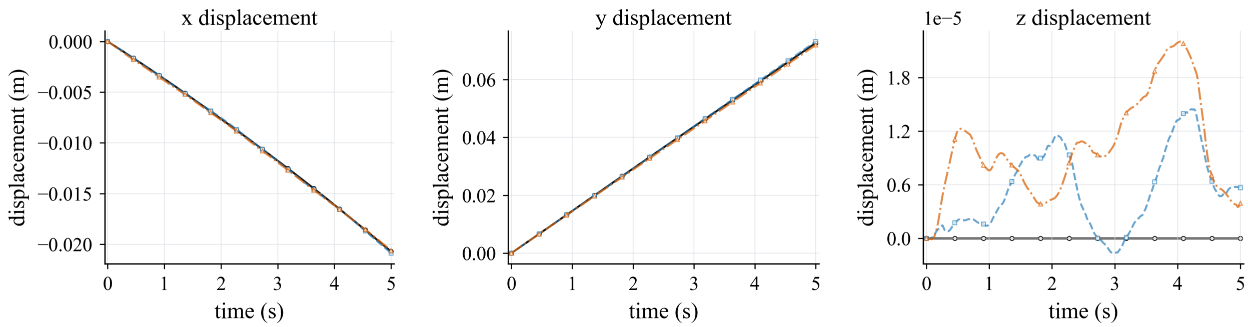

Figure 19: Bearing raceway validation over 5 s: ball displacement in the x, y, and z directions. The CALG result is shown as the black solid curve; displacements are relative to each solver's initial ball coordinate.

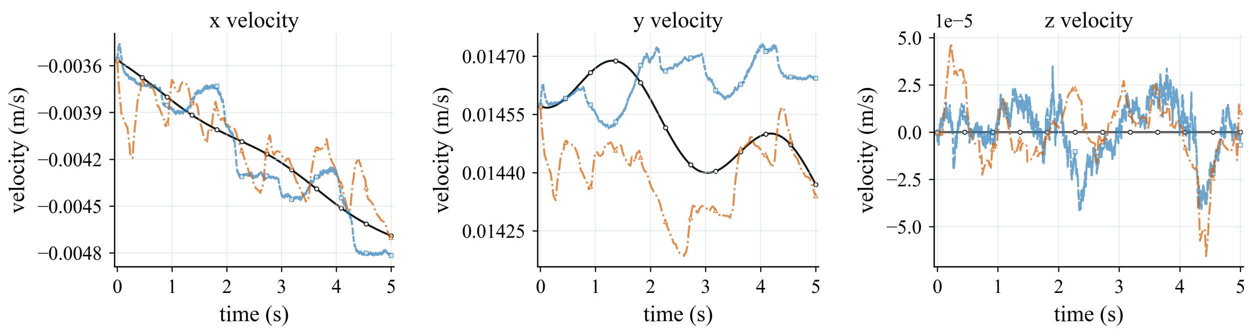

Figure 20: Bearing raceway validation over 5 s: ball velocity in the x, y, and z directions. The CALG result is shown as the black solid curve.

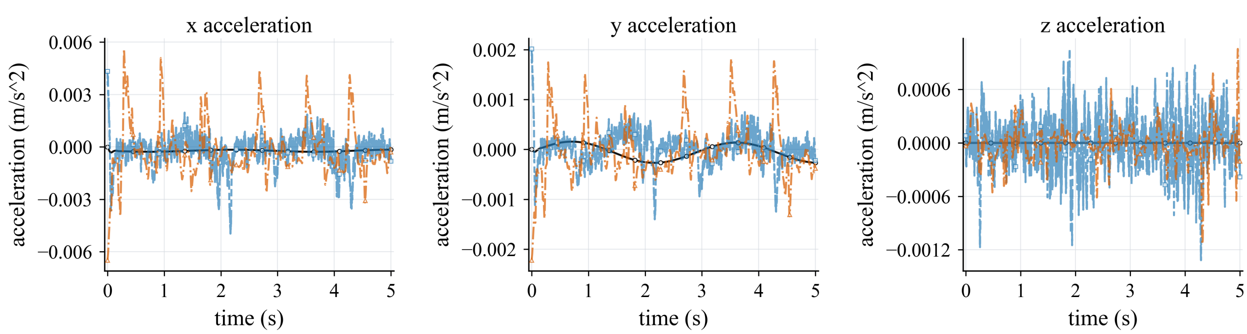

Figure 21: Bearing raceway validation over 5 s: ball acceleration in the x, y, and z directions. For CALG and Recurdyn, acceleration is taken from the exported solver channels; for Chrono it is evaluated from the reported contact force. The plotted curves use a 41-frame centered average for visualization.

Together, the three external-validation scenes show that the same CALG–SDF contact samples can drive consistent rigid-motion trends in selected engineering settings while preserving the smoother contact-normal and friction-response behavior established in the direct backend comparison.

Table 5 connects the external-validation scenes back to the paper's contact-geometry mechanism by replaying the selected rigid states with the three geometry modules. This geometry-backend replay diagnostic is not an external-solver error table; it changes only the contact-geometry module along the same trajectory. The guide slot and ball-and-socket pendulum diagnostics reproduce the expected Linear-triangle BVH failure mode: the facet-based module causes much larger normal and normal-force jumps under the same friction parameters. The bearing raceway diagnostic is intentionally retained as a neutral control: the locked bearing-raceway trajectory spans only a small angular interval, so it is not a discriminating normal-jump test, but it verifies that the three contact-geometry evaluations remain comparable in a smooth near-steady raceway segment.

4.6. C++ backend efficiency

The Python engineering suite is useful for development but not an appropriate efficiency baseline. To measure module cost more fairly, all three selected validation scenes were run through the same native CALG C++ backend and compared with Chrono wall-clock time and Recurdyn solver-reported CPU Time. Table 6 reports CALG solver-loop time without full CSV trace export and excludes one-time geometry preprocessing such as dense SDF construction. In all three scenes, the reported CALG timing uses local mesh/CALG detector traversal followed by configuration-space SDF refinement and normalized surface-contact response quadrature for the signed gap, normal, and frictional response, rather than a faster analytic-provider-only replay.

Figure 22 visualizes the same solver-timing data. These values should be read as implemented validation benchmark timings, not as a general engine-level comparison: Chrono, Recurdyn, and CALG use different contact discretizations and regularizations. Within this benchmark, CALG reports lower solver time than Chrono in all three selected scenes.

Table 5: Geometry-backend replay diagnostic on the three external-validation trajectories. The locked CALG detector trajectory is replayed and the contact-geometry module is changed while keeping the rigid state and friction parameters fixed.

Model	Metric	Linear-triangle BVH	PN-triangle BVH	CALG
Guide slot	$\Delta n_{\max}$ (deg)	17.98	1.04	0.945
	$\Delta F_{n,\max}$ (N)	302.09	245.44	261.20
	$\Delta F_{t,\max}$ (N)	11.98	11.33	10.96
	v_t jitter (m/s)	2.25×10^{-2}	2.69×10^{-3}	2.74×10^{-3}
Ball-and-socket pendulum	$\Delta n_{\max}$ (deg)	4.67	0.0618	0.0176
	$\Delta F_{n,\max}$ (N)	0.562	0.0467	0.0467
	$\Delta F_{t,\max}$ (N)	0.0229	0.00415	0.00415
	v_t jitter (m/s)	1.34×10^{-4}	1.34×10^{-4}	1.33×10^{-4}
Bearing raceway	$\Delta n_{\max}$ (deg)	0	0	1.63×10^{-3}
	$\Delta F_{n,\max}$ (N)	0.506	0.504	0.505
	$\Delta F_{t,\max}$ (N)	1.26×10^{-4}	5.38×10^{-5}	5.38×10^{-5}
	v_t jitter (m/s)	6.33×10^{-5}	6.15×10^{-5}	6.16×10^{-5}

Table 6: Solver timing for Chrono, Recurdyn, and CALG. Chrono values are wall-clock times for the corresponding solver calls; Recurdyn values are solver-reported CPU Time; CALG values are native C++ solver-loop times and exclude one-time geometry preprocessing such as dense SDF construction. The Time/CALG rows report each solver time divided by the CALG time for the same case. Host hardware: AMD Ryzen 9 5900X, 12 cores/24 threads, Windows 11 Pro 64-bit.

Case	Metric	Chrono	Recurdyn	CALG
Guide slot	Time (s)	0.87	12.40	0.58
	Time/CALG	1.50×	21.36×	1.00×
Ball-and-socket pendulum	Time (s)	150.22	454.83	28.10
	Time/CALG	5.35×	16.19×	1.00×
Bearing raceway	Time (s)	49.43	61.19	20.17
	Time/CALG	2.45×	3.03×	1.00×

Figure 22: Solver timing per selected validation scene. CALG uses no-CSV solver-loop time and excludes one-time geometry preprocessing such as dense SDF construction; Recurdyn uses solver-reported CPU Time. All three CALG rows include local mesh/CALG detector traversal, configuration-space SDF signed refinement, and normalized surface-contact response quadrature. Timing ratios are reported in Table 6 and describe this implemented validation benchmark only, not a solver-wide engine comparison.

4.7. Scope of the current evidence

The current evidence supports a specific claim: for smooth engineering contact, the CALG–SDF pipeline provides smoother signed gaps, normals, and tangential frames for rigid frictional response than fixed faceted or fixed-tessellation contact-geometry modules. The executable results in this paper use rigid-body contact samples, detector-driven candidate selection, configuration-space SDF signed refinement in the C++ validation backend, and regularized Coulomb friction. This distinction is important for interpreting the method as a contact-geometry backend rather than as a solver-wide replacement for established multibody dynamics engines.

5. Conclusion

This paper addresses frictional contact artifacts caused by discontinuous or tessellation-dependent contact geometry. The key idea is to couple a contact-aligned local graph detector with a configuration-space signed-distance response layer. The local graph supplies the signed gap, contact normal, and tangent frame from the local contact geometry, while the SDF layer provides the smooth signed gap and gradient normal used by the rigid friction law.

The reported results support the method at three levels. Analytic inclined-plane controls verify that the rigid-friction update reproduces static sticking, kinetic sliding, and the expected threshold transition. Dynamic rigid-friction tests then show that Linear-triangle BVH and PN-triangle BVH produce substantially larger normal jumps, normal-force jumps, friction-force jumps, and tangential-speed jitter than CALG in the same four engineering scenes. Chrono and Recurdyn comparisons confirm trajectory-level agreement in displacement and velocity trends in three selected engineering scenes. In the C++ timing benchmark, the CALG–SDF backend reports lower solver-loop time than Chrono in all three selected scenes, with the comparison limited by different contact regularizations.

The present limitation is scope rather than the central geometric mechanism. The implementation and validation focus on rigid frictional contact over smooth analytic or CAD-like patches, with a configuration-space SDF used as the signed-gap response layer in the final C++ backend. Sparse or adaptive response-field storage and production-grade multiphysics integration remain future work.

CRediT author statement

Anonymous Author(s): Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, and Visualization.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The code, input definitions, scripts, generated figures, and numerical result artifacts required to reproduce the reported tables are available in the public repository at `https://github.com/W-Moorer/bff-rigid-contact.git`. The result files used in this draft are stored in the repository result directories v0.6, v0.9, v0.10, v0.11, v0.12, v0.13, v0.14, and v0.15. The repository includes source-result mappings, repository metadata, and a file-level manifest with SHA256 checksums.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI Codex to assist with manuscript restructuring, wording, consistency checks, and preparation of reproducibility and plotting scripts. No generative image model was used to create the scientific figures. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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